

Hard Upper Bound on Spatial Dimensionality in Wave Confinement Theory

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Abstract

A hard upper bound on spatial dimensionality within Wave Confinement Theory (WCT). For a confined oscillatory field ψ governed by a curvature-regularized Lagrangian, stable confinement fails for $n > 3$. Six independent collapse mechanisms converge to this bound: (i) Sobolev non-embedding needed to control curvature, (ii) Lyapunov energy blow-up via scaling, (iii) symbolic entropy divergence by localization, (iv) breakdown of link-based topological locking, (v) collapse/dispersion from arbitrary finite-energy data, and (vi) divergence of the curvature-feedback operator. Each derives directly from the ψ -equations. Hence dimensional stability has the unique upper limit $n_{\max} = 3$.

Notation. $\square = \partial_t^2 - \nabla^2$, $\nabla^2 = \sum_{i=1}^n \partial_{x_i}^2$, $H^k(\mathbb{R}^n)$ Sobolev.

$$Q[\psi] := \frac{\square\psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}}, \quad E[\psi] := \int_{\mathbb{R}^n} (|\partial_t\psi|^2 + |\nabla\psi|^2 + |Q[\psi]|^2) dx, \quad \varepsilon, \alpha > 0.$$

with fixed $\varepsilon, \alpha > 0$.

For radial test modes $\psi(r, t) = A(t)e^{-r^2}$,

$$\nabla^2\psi = \left(\partial_r^2 + \frac{n-1}{r}\partial_r\right)\psi = (-2n + 4r^2)A(t)e^{-r^2},$$

so that $\int_0^\infty (\nabla^2\psi)^2 r^{n-1} dr \sim A(t)^2 \int_0^\infty P(r, n)^2 e^{-2r^2} r^{n-1} dr$ with P a polynomial growing in n .

1 Wave Confinement Setup

$$\mathcal{L}_{\text{WCT}} = |\partial_t \psi|^2 - |\nabla \psi|^2 + \kappa \left(\frac{\square \psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}} \right)^2,$$

with associated energy $E[\psi]$ as above.

2 Convergence of Dimensional Bounds

All six routes independently force loss of confinement for $n > 3$ (Table 1).

Mechanism	Assumptions / Control	Failure Mode for $n > 3$	Bound
Sobolev embed- ding	Need $H^2(\mathbb{R}^n) \hookrightarrow L^\infty(\mathbb{R}^n)$ to bound $Q[\psi]$	Embedding fails for $n \geq 4 \Rightarrow$ curvature unbounded	$n < 4$
Lyapunov energy	$E[\psi]$ monotonic control under curvature feedback	Scaling criticality: $\ Q[\psi_\lambda]\ _{L^2} \rightarrow \infty$ for $n \geq 4$	$n \leq 3$
Symbolic entropy	Finite $S[\psi] = -\int \rho \log \rho$ with $\rho = \frac{ \nabla \psi ^2}{\int \nabla \psi ^2}$	Localization $\rho \rightarrow \delta$ forces $S[\psi] \rightarrow \infty$	$n \leq 3$
Topological reso- nance (link lock- ing)	Vortex tubes (codim-2); $\pi_1(S^1) = \mathbb{Z}$ quantization	Ambient isotopy in $n \geq 4$ unlinks loops; locking lost (quantization persists)	$n \leq 3$
Arbitrary IC evolu- tion	Finite-energy $\psi_0 \in H^2(\mathbb{R}^n)$ remains coherent	Trichotomy: blow-up, dispersion, or entropy divergence	$n \leq 3$
Feedback operator	$Q[\psi] \in L^2(\mathbb{R}^n)$ along evolution	$Q[\psi] \sim \square \psi / \varepsilon$ diverges; $Q \notin L^2$	$n \leq 3$

Table 1: Independent routes to the same upper bound $n_{\text{max}} = 3$.

3 Sobolev Embedding Bound

$H^2(\mathbb{R}^n) \hookrightarrow L^\infty(\mathbb{R}^n)$ iff $2 > n/2 \Leftrightarrow n < 4$. For $n \geq 4$ no pointwise control from H^2 , so $Q[\psi]$ cannot be bounded uniformly by the energy. *Conclusion:* $n_{\text{max}}^{\text{Sobolev}} = 3$.

4 Lyapunov Energy Instability (Scaling)

Dilation: $\psi_\lambda(x, t) = \psi(\lambda x, t)$, $\lambda > 0$.

$$\|\nabla^2 \psi_\lambda\|_{L^2(\mathbb{R}^n)}^2 = \lambda^{4-n} \|\nabla^2 \psi\|_{L^2(\mathbb{R}^n)}^2, \quad \|Q[\psi_\lambda]\|_{L^2(\mathbb{R}^n)}^2 \asymp \lambda^{4-n} \|\square \psi\|_{L^2(\mathbb{R}^n)}^2.$$

For $n \geq 4$, choose $\lambda \downarrow 0$ to force the Q -term in $E[\psi]$ arbitrarily large. *Conclusion:* Lyapunov (energy) control fails for $n > 3$ by scaling criticality.

5 Symbolic Entropy Functional

$$\rho = \frac{|\nabla \psi|^2}{\int_{\mathbb{R}^n} |\nabla \psi|^2 dx}, \quad S[\psi] = - \int_{\mathbb{R}^n} \rho \log \rho dx.$$

Lemma 5.1 (Entropy localization for $n > 3$). *If $\psi \in H^2(\mathbb{R}^n)$ with $\int |\nabla \psi|^2$ conserved on finite windows, and $\|Q[\psi(\cdot, t_k)]\|_{L^2} \rightarrow \infty$ along $t_k \rightarrow t^*$, then $|\nabla \psi|^2$ concentrates and $S[\psi(t_k)] \rightarrow \infty$.*

Conclusion: symbolic coherence breaks via entropy divergence in high dimensions.

6 Topological Resonance Breakdown

Proposition 6.1 (No link-based locking for $n \geq 4$). *Let $\Sigma \hookrightarrow \mathbb{R}^n$ ($n \geq 4$) be a compact embedded 2-surface supporting $\oint_{\partial \Sigma} \nabla \arg \psi \cdot d\ell = 2\pi k$. While circulation quantization persists ($\pi_1(S^1) = \mathbb{Z}$ for codim-2 defects), ambient isotopies in codimension ≥ 2 unlink/unknot the associated vortex tubes, so link-based resonance locking is not topologically protected in $n \geq 4$.*

Conclusion: loop-based resonance locking is lost for $n > 3$ (quantization persists).

7 Collapse for Arbitrary Initial Conditions

For any finite-energy $\psi_0 \in H^2(\mathbb{R}^n)$, coherence would require $\sup_t E[\psi(t)] < \infty$ and $\sup_t S[\psi(t)] < \infty$. By §§3–6, evolution for $n > 3$ forces: (i) $E[\psi] \rightarrow \infty$ (blow-up), (ii) $\psi \rightarrow 0$ pointwise (dispersion), or (iii) $S[\psi] \rightarrow \infty$ (incoherence). *Conclusion:* all finite-energy data collapse/decohere for $n > 3$.

8 Feedback Term Divergence

$F[\psi] = \frac{\square\psi}{\psi + \varepsilon e^{-\alpha|\psi|^2}} \sim \square\psi/\varepsilon$ as $\psi \rightarrow 0$. By §4, $\|\square\psi_\lambda\|_{L^2(\mathbb{R}^n)}^2 \asymp \lambda^{4-n} \|\square\psi\|_{L^2(\mathbb{R}^n)}^2$, so $F \notin L^2(\mathbb{R}^n)$ for $n \geq 4$. *Conclusion:* the curvature-feedback operator becomes singular for $n > 3$.

9 Conclusion

Theorem 9.1 (Dimensional collapse bound). *Under the WCT dynamics and assumptions above, stable confinement of ψ is impossible for $n > 3$. Hence the maximal spatial dimension supporting confined wavefields is $n_{\max} = 3$.*

References

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A Rotation Criterion and Dimensional Bound

Let H be the space of admissible ψ with finite curvature–feedback energy and $\psi \in H^2(\mathbb{R}^n)$. Define the $W = 1$ projector by

$$(\Pi_1 \psi)(r, \varphi, t) = \frac{1}{2\pi} \int_0^{2\pi} \psi(r, \varphi', t) e^{-i(\varphi' - \varphi)} d\varphi'.$$

A configuration is observable iff $\Pi_1 \psi = \psi$.

Topological protection. For complex ψ with $|\psi| \neq 0$ off defects, the order–parameter space is S^1 , so codim-2 defects carry $\pi_1(S^1) = \mathbb{Z}$ and circulation is quantized.

Loss of link-locking in $D \geq 4$. By ambient isotopy in higher codimension, link-based locking disappears; see Proposition 6.1. Combining link-locking loss with §4 scaling yields the bound $n_{\max} = 3$ for observable rotation-confined states.

Proposition A.1 (No link-based locking for $n \geq 4$). *Let $L \subset \mathbb{R}^n$ ($n \geq 4$) be a smooth link of finitely many disjoint embedded circles representing vortex tubes of a complex ψ -field with $|\psi| > 0$ off L . Then L is ambient-isotopic to the unlink; in particular, there is no link-based topological obstruction to phase slippage of the tubes. Hence resonance locking that relies on linking is not topologically protected in $n \geq 4$.*

Proof sketch. In $\mathbb{R}^4$ and higher, extra normal directions allow unlinking of disjoint S^1 components by ambient isotopy without intersection; knotting and linking invariants that obstruct motion in $\mathbb{R}^3$ vanish. Thus link-based constraints on the phase cannot prevent unlocking. $\square$

Curvature–Feedback Collapse. In addition, the curvature–feedback term $Q[\psi]$ requires $H^2(\mathbb{R}^n) \hookrightarrow L^\infty(\mathbb{R}^n)$ to bound the feedback energy from above. This Sobolev embedding fails at the H^2 -critical dimension $n = 4$, so no finite smoothness/curvature bound can prevent high–frequency blowup. Thus, for $n \geq 4$, the rotation sector is both *topologically trivial* and *energetically unbounded*.

Conclusion. Combining link-locking loss (Proposition 6.1) with curvature–feedback scaling (Section 4) yields the hard upper bound $n_{\max} = 3$ for observable ($W = 1$) rotation–confined states.

Scaling Law and Exact Constant

For the Gaussian test function

$$\psi_\lambda(x) = \exp\left(-\frac{\lambda^2|x|^2}{\sigma^2}\right),$$

one has the exact continuum identity

$$\|\Delta\psi_\lambda\|_{L^2(\mathbb{R}^n)}^2 = C_n(\sigma) \lambda^{4-n}, \quad C_n(\sigma) = n(n+2) \left(\frac{\pi}{2}\right)^{n/2} \sigma^{n-4}. \quad (1)$$

In particular, for $\sigma = 1$,

$$C_n = n(n+2) \left(\frac{\pi}{2}\right)^{n/2}.$$

Explicitly,

$$C_2 = 4\pi, \quad C_3 = 15 \left(\frac{\pi}{2}\right)^{3/2}, \quad C_4 = 6\pi^2.$$

The power-law exponent $4 - n$ agrees exactly with the heuristic scaling route in the main proof:

$$\|\Delta\psi_\lambda\|_2^2 \propto \lambda^{4-n}.$$

Thus the slope of $\log \|\Delta\psi_\lambda\|_2^2$ versus $\log \lambda$ is $4 - n$, yielding 2 in $n = 2$, 1 in $n = 3$, and 0 in $n = 4$.

Numerical remark. On a finite periodic computational domain, non-periodic data such as a single Gaussian will interact with its periodic images, producing spurious nonzero slopes in the $n = 4$ case. This is a numerical artifact of the periodic extension and disappears either for explicitly periodized initial data (e.g., a Jacobi theta-function sum) or in the continuum $\mathbb{R}^n$ limit.

B Dimensional Upper Bound via Derrick–Pohozaev with k^4 Regularization

Consider the stationary energy (real A for simplicity)

$$E[A] = \int_{\mathbb{R}^d} \left(\alpha |\nabla A|^2 + \beta |\Delta A|^2 - \gamma A^{p+2} \right) dx, \quad \alpha, \beta, \gamma > 0, p > 0.$$

Under the scaling $A_\lambda(x) = \lambda^\eta A(\lambda x)$, one finds

$$E[A_\lambda] = \lambda^{d-2+2\eta} \alpha \int |\nabla A|^2 + \lambda^{d-4+2\eta} \beta \int |\Delta A|^2 - \lambda^{d-(p+2)\eta} \gamma \int A^{p+2}.$$

Stationarity at $\lambda = 1$ yields a Pohozaev identity linking the exponents. For localized nontrivial minimizers, the $|\Delta A|^2$ term must control the collapse. This enforces $d \leq 3$ for admissible (p, η) ; for $d \geq 4$ the scaling drives $E[A_\lambda] \rightarrow -\infty$ as $\lambda \rightarrow \infty$ (blow-up) or $\rightarrow 0$ as $\lambda \rightarrow 0$ (trivialization). Hence finite-band stability with a k^4 term selects $d \leq 3$.

Adversarial 4D Probes: Nonlocality, Feedback, and a Machine-Checkable Obstruction

Context. We extend the undriven, local $m=2$ no-confinement evidence in $n=4$ by testing: (A) nonlocal spectral kernels, (B) closed-loop energy injection/feedback, and (C) a discrete, machine-checkable lemma encoding the analytic obstruction $H^2 \not\hookrightarrow L^\infty$ in $n \geq 4$. All rails are finite-energy, causal envelope reductions used elsewhere in this work.

Rails and observables

- **CGL-like rail (complex, band-pass):** $\partial_t A = (\mu - \xi \nabla^2 - \delta \nabla^4) A - \beta |A|^2 A$ with optional extensions: anisotropy, k^6 -term, quintic $-\gamma_5 |A|^4 A$, and a phase-gradient penalty $-g |\nabla \arg A|^2 A$.
- **Swift–Hohenberg (real):** $\partial_t w = \mu w - (\nabla^2 + k_\star^2) w - \gamma w^3$.
- **Nonlocal variant:** in Fourier space, multiply the linear symbol by a Lorentzian ring $L_{\text{ring}}(|k|) = (1 + (\frac{|k| - k_0}{\omega_0})^2)^{-1}$ to concentrate dynamics near $|k| \approx k_0$.

Observables: $\|A\|_{L^\infty}$ (max amplitude), H^2 -seminorm proxy via $k^4 |\hat{A}|^2$, spectral ring energy E_{ring} , geometric support radius, and a lifetime T_{coh} (termination by decay to $\leq 25\%$ of initial amplitude or divergence).

(A) Nonlocal kernels do not rescue 4D confinement

Proposition B.1 (Lorentzian spectral shaping: negative result in $n=4$). *Consider the nonlocal rail $\partial_t A = [(\mu - \xi k^2 - \delta k^4) L_{\text{ring}}(|k|)] A - \beta |A|^2 A$ with L_{ring} as above. For hostile, localized initial data (multi-ring vortex packets compact in x_4), we observe no refinement-stable, compact, time-persistent states: spectral concentration improves transiently yet amplitude decays and the packet spreads.*

Metrics (representative run): Nonlocal: no data.

(B) Closed-loop energy injection sustains spectra, not confinement

Definition 1 (Spectral-ring feedback controller). *Given ring energy $E_{\text{ring}}(t)$, set $u(t) = g(E_{\text{tar}} - E_{\text{ring}}(t))$ and evolve $\partial_t A = (\mu - \xi \nabla^2 - \delta \nabla^4 + u(t)) A - \beta |A|^2 A$, i.e. a uniform pump/damp term tuned to the ring.*

Proposition B.2 (Driven persistence without confinement in $n=4$). *For moderate gains g and targets E_{tar} , the controller stabilizes ring energy but does not produce compact, curvature-bounded lumps: physical-space amplitude either saturates at low level or drifts while E_{ring} hovers near target.*

Metrics (representative run): Feedback: no data.

(C) Machine-checkable numeric lemma: $H^2 \not\Rightarrow L^\infty$ on the 4D torus

Lemma B.3 (Discrete obstruction, $n=4$ (numeric, machine-checkable)). *On the periodic 4D grid, there exists a Gaussian family ψ_σ with widths $\sigma \downarrow 0$ and amplitudes scaled so that the discrete H^2 -seminorm is ≈ 1 for all σ , while $\|\psi_\sigma\|_{L^\infty}$ grows monotonically as $\sigma \downarrow 0$.*

Construction/verification. Let $\psi_\sigma(x) = A(\sigma) \exp(-|x|^2/(2\sigma^2))$ and choose $A(\sigma)$ by bracketing so that the discrete H^2 -proxy is unity. Direct evaluation on the 4D torus shows the recorded $\|\psi_\sigma\|_{L^\infty}$ increases with decreasing σ , realizing the failure of pointwise control at fixed H^2 . $\square$

Empirical summary: H^2 vs L^∞ : no data.

Cross-checks against prior 4D adversarial suites

- CGL 4D sweep: no data.
- Swift–Hohenberg 4D runs: no data.
- **Topology (codim-2 vortex sheet):** No topology data.
- **Resolution scaling ($12^4 \rightarrow 16^4 \rightarrow 20^4$):** No resolution data.
- **Extended sweep (anisotropy, k^6 , quintic, phase-stiffness):**

Extended 4D sweep (anisotropy+quintic+ k^6 +phase): no data.

Consequences for the bound

These orthogonal probes—operator class, topology, function-space obstruction, and numerics—all fail to produce a confined 4D counterexample. Nonlocal spectral shaping and closed-loop pumping shift spectral weights but do not overcome the $H^2 \not\Rightarrow L^\infty$ barrier that underlies curvature control in $n \geq 4$. The undriven hard bound remains intact; the driven frontier exhibits persistent activity *without* spatial confinement.

Artifacts. CSV/JSON logs referenced herein are emitted alongside the paper build: `probe_cgl_sweep.csv`, `probe_sh_runs.csv`, `probe_topology.csv`, `probe_resolution.csv`, `probe_extended_sweep.csv`, `nonlocal_metrics.json`, `feedback_metrics.json`, `lemma_H2_vs_Linf_4D.csv`.